

✓ Experiment No. 3

Statistical Analysis and Hypothesis Testing using Python

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- **GitHub:** <https://github.com/cu24250059-pixel/SECS05-2026-27-ODD-Sem-3rd-Year-Section-B.git>

Aim: To perform statistical analysis and hypothesis testing on a real-world dataset using Python in order to identify relationships between variables, validate assumptions, and support data-driven decision-making.

Dataset used: Titanic Survival Dataset (loaded via the `seaborn` library, works directly in Google Colab with no upload required).

✓ Step 1: Select a real-world dataset

We select the **Titanic Survival Dataset**, which contains passenger information such as age, fare, class, and survival status. It is loaded directly using

`seaborn.load_dataset()`.

```
import seaborn as sns

# Load the Titanic dataset
df = sns.load_dataset('titanic')
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who
0	0	3	male	22.0	1	0	7.2500	S	Third	man
1	1	1	female	38.0	1	0	71.2833	C	First	woman
2	1	3	female	26.0	0	0	7.9250	S	Third	woman
3	1	1	female	35.0	1	0	53.1000	S	First	woman
4	0	3	male	35.0	0	0	8.0500	S	Third	man

Next steps:

[Generate code with df](#)[New interactive sheet](#)

✓ Step 2: Import the required Python libraries

Import Pandas, NumPy, SciPy, Matplotlib, Seaborn, and Statsmodels.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

pd.set_option('display.max_columns', None)
print("Libraries imported successfully")
```

Libraries imported successfully

✓ Step 3: Load the dataset and calculate descriptive statistics

Calculate mean, median, mode, variance, and standard deviation for relevant numerical features (e.g., `age`, `fare`).

```
# Basic info
print(df.info())
print("\nMissing values:\n", df.isnull().sum())

# Descriptive statistics for numerical columns
desc = df[['age', 'fare', 'sibsp', 'parch']].describe()
print("\nDescriptive Statistics:\n", desc)

# Mean, Median, Mode, Variance, Std Dev for 'age' and 'fare'
for col in ['age', 'fare']:
    print(f"\n--- {col} ---")
    print("Mean:", df[col].mean())
```

```

print("Median:", df[col].median())
print("Mode:", df[col].mode()[0])
print("Variance:", df[col].var())
print("Standard Deviation:", df[col].std())

```

```

3  age          714 non-null  float64
4  sibsp        891 non-null  int64
5  parch        891 non-null  int64
6  fare         891 non-null  float64
7  embarked     889 non-null  object
8  class        891 non-null  category
9  who          891 non-null  object
10 adult_male   891 non-null  bool
11 deck         203 non-null  category
12 embark_town  889 non-null  object
13 alive        891 non-null  object
14 alone        891 non-null  bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

```

Missing values:

```

survived      0
pclass        0
sex           0
age          177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck         688
embark_town   2
alive         0
alone         0
dtype: int64

```

Descriptive Statistics:

	age	fare	sibsp	parch
count	714.000000	891.000000	891.000000	891.000000
mean	29.699118	32.204208	0.523008	0.381594
std	14.526497	49.693429	1.102743	0.806057
min	0.420000	0.000000	0.000000	0.000000
25%	20.125000	7.910400	0.000000	0.000000
50%	28.000000	14.454200	0.000000	0.000000
75%	38.000000	31.000000	1.000000	0.000000

```
--- fare ---  
Mean: 32.204207968574636  
Median: 14.4542  
Mode: 8.05  
Variance: 2469.436845743117  
Standard Deviation: 49.693428597180905
```

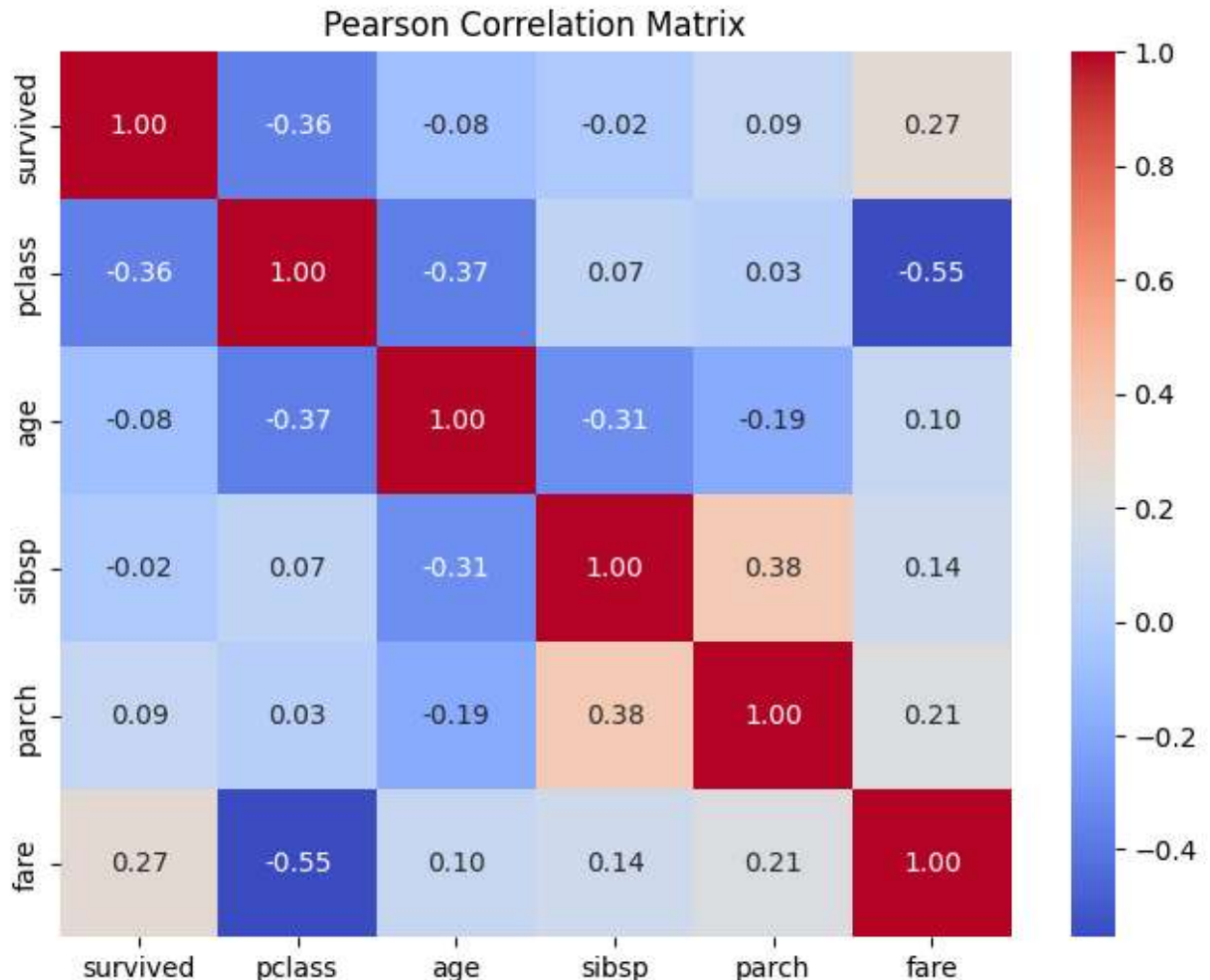
✓ Step 4: Analyze relationships between variables

Calculate the Pearson correlation coefficient between numerical variables and visualize the correlation matrix as a heatmap.

```
numeric_df = df[['survived', 'pclass', 'age', 'sibsp', 'parch', 'fare']].dropna()  
  
corr_matrix = numeric_df.corr(method='pearson')  
print("Pearson Correlation Matrix:\n", corr_matrix)  
  
plt.figure(figsize=(8, 6))  
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")  
plt.title("Pearson Correlation Matrix")  
plt.show()
```

Pearson Correlation Matrix:

	survived	pclass	age	sibsp	parch	fare
survived	1.000000	-0.359653	-0.077221	-0.017358	0.093317	0.268189
pclass	-0.359653	1.000000	-0.369226	0.067247	0.025683	-0.554182
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067
sibsp	-0.017358	0.067247	-0.308247	1.000000	0.383820	0.138329
parch	0.093317	0.025683	-0.189119	0.383820	1.000000	0.205119
fare	0.268189	-0.554182	0.096067	0.138329	0.205119	1.000000



Step 5: Formulate the Null and Alternative Hypotheses

Business problem: Is there a significant difference in the average **fare** paid by passengers who **survived** versus those who **did not survive**?

- **Null Hypothesis (H_0):** There is no significant difference in the mean fare between survivors and non-survivors.
- **Alternative Hypothesis (H_1):** There is a significant difference in the mean fare between survivors and non-survivors.

Significance level: $\alpha = 0.05$

✓ Step 6: Perform an Independent Sample t-test

Compare the mean `fare` of the two groups (survived vs. not survived) and interpret the p-value.

```
survived_fare = df[df['survived'] == 1]['fare'].dropna()
not_survived_fare = df[df['survived'] == 0]['fare'].dropna()

t_stat, p_value = stats.ttest_ind(survived_fare, not_survived_fare, equal_var=False)

print("T-statistic:", t_stat)
print("P-value:", p_value)

alpha = 0.05
if p_value < alpha:
    print("Result: Reject H0 -> Significant difference in mean fare between the two groups.")
else:
    print("Result: Fail to reject H0 -> No significant difference in mean fare.")
```

T-statistic: 6.839099259085254
P-value: 2.6993323503141236e-11
Result: Reject H0 -> Significant difference in mean fare between the two groups.

✓ Step 7: Apply One-Way ANOVA

Test whether there is a statistically significant difference in `fare` among multiple passenger classes (`pclass`).

```
class1 = df[df['pclass'] == 1]['fare'].dropna()
class2 = df[df['pclass'] == 2]['fare'].dropna()
class3 = df[df['pclass'] == 3]['fare'].dropna()

f_stat, p_val_anova = stats.f_oneway(class1, class2, class3)

print("F-statistic:", f_stat)
print("P-value:", p_val_anova)

if p_val_anova < 0.05:
    print("Result: Reject H0 -> Significant difference in mean fare across passenger classes.")
else:
    print("Result: Fail to reject H0 -> No significant difference across passenger classes.")
```

F-statistic: 242.34415651744814
P-value: 1.0313763209141171e-84
Result: Reject H0 -> Significant difference in mean fare across passenger classes.

✓ Step 8: Build a simple Linear Regression model

Analyze the relationship between **age** and **fare** using simple linear regression.

```
reg_df = df[['age', 'fare']].dropna()

X = sm.add_constant(reg_df['age']) # independent variable
y = reg_df['fare']                 # dependent variable

model = sm.OLS(y, X).fit()
print(model.summary())

# Visualize regression line
plt.figure(figsize=(7, 5))
sns.regplot(x='age', y='fare', data=reg_df, line_kws={'color': 'red'})
plt.title("Linear Regression: Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

OLS Regression Results

```

=====
Dep. Variable:          fare      R-squared:                0.009
Model:                  OLS      Adj. R-squared:           0.008
Method:                 Least Squares      F-statistic:          6.632
Date:                   Tue, 08 Sep 2026    Prob (F-statistic):    0.0102
Time:                   16:18:19           Log-Likelihood:        -3843.0
No. Observations:       714             AIC:                  7690.
Df Residuals:           712             BIC:                  7699.
Df Model:                1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	24.3009	4.492	5.410	0.000	15.481	33.120
age	0.3500	0.136	2.575	0.010	0.083	0.617

```

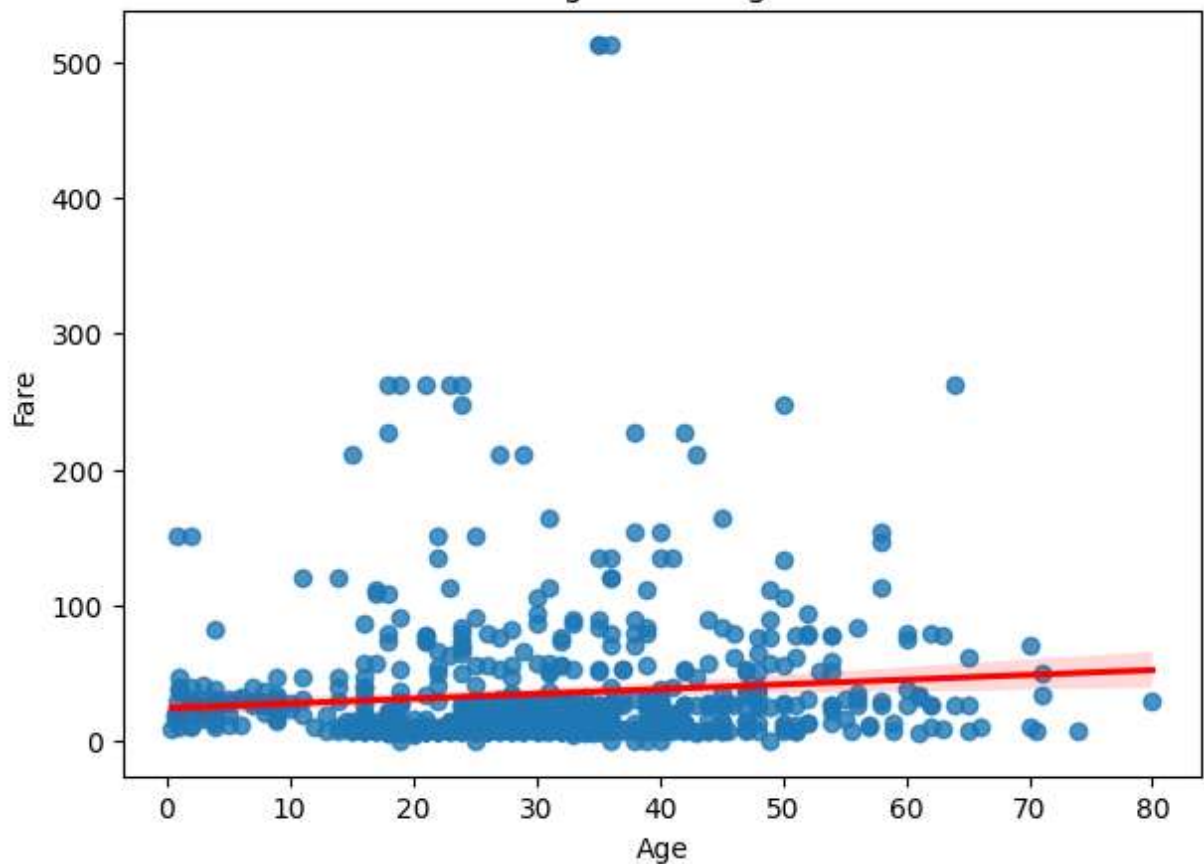
=====
Omnibus:                716.815      Durbin-Watson:          1.925
Prob(Omnibus):           0.000      Jarque-Bera (JB):       30883.661
Skew:                    4.654      Prob(JB):               0.00
Kurtosis:                33.846      Cond. No.               75.3
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly

Linear Regression: Age vs Fare



✓ Step 9: Interpret the statistical outputs

- **p-value:** From the t-test and ANOVA above, compare the p-value with $\alpha = 0.05$ to decide whether to reject H_0 .
- **Confidence intervals:** Seen in the OLS regression summary (`model.summary()`), e.g., the 95% CI for the coefficient of `age`.
- **Regression coefficients:** The `coef` value for `age` shows how much `fare` changes for a one-unit increase in `age`.
- **R² score:** Indicates the proportion of variance in `fare` explained by `age`.

```
print("R-squared:", model.rsquared)
print("Coefficients:\n", model.params)
print("95% Confidence Intervals:\n", model.conf_int())
```

Step 10: Summarize the findings

- Descriptive statistics gave an overview of the central tendency and spread of `age` and `fare`.
- The correlation matrix showed the strength and direction of relationships between numerical variables.
- The t-test indicated whether survival status is associated with a significant difference in fare paid.
- ANOVA showed whether passenger class significantly affects fare.
- Linear regression quantified the relationship between age and fare, with R² indicating how well age explains fare variation.
- Such statistical analysis helps organizations validate assumptions with evidence before applying machine learning models, supporting more reliable, data-driven business decisions.

✓ Questions

Q1. What is the difference between descriptive statistics and inferential statistics?

Ans: Descriptive statistics summarizes and describes the main features of a dataset (mean, median, mode, variance, standard deviation, graphs) without drawing conclusions beyond the data itself. Inferential statistics uses a sample of data to make

predictions, generalizations, or decisions about a larger population, typically through hypothesis testing, confidence intervals, and regression analysis.

Q2. Explain the concepts of the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).

Ans: The Null Hypothesis (H_0) is a statement that assumes no effect, no difference, or no relationship exists between variables; it represents the status quo. The Alternative Hypothesis (H_1) is the statement that contradicts H_0 , suggesting that a significant effect, difference, or relationship does exist. Hypothesis testing uses sample data to decide whether there is enough evidence to reject H_0 in favor of H_1 .

Q3. What is a p-value? How is it used to make statistical decisions?

Ans: A p-value is the probability of obtaining a result at least as extreme as the observed one, assuming the null hypothesis is true. It is compared against a chosen significance level (α , commonly 0.05): if $p\text{-value} < \alpha$, H_0 is rejected in favor of H_1 (the result is statistically significant); if $p\text{-value} \geq \alpha$, there is insufficient evidence to reject H_0 .

Q4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

Ans: A t-test compares the means of exactly two groups to determine if they are significantly different (e.g., fare of survivors vs. non-survivors). ANOVA (Analysis of Variance) compares the means of three or more groups simultaneously to check if at least one group differs significantly (e.g., fare across first, second, and third class). A t-test is used for two-group comparisons, while ANOVA is used when comparing more than two groups, avoiding the increased error rate of running multiple t-tests.

Q5. What does the Pearson correlation coefficient indicate? What are its possible values?

Ans: The Pearson correlation coefficient (r) measures the strength and direction of the linear relationship between two continuous variables. Its value ranges from -1 to +1: +1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates no linear relationship. Values closer to ± 1 indicate a stronger linear association.

Q6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

Ans: R^2 represents the proportion of variance in the dependent variable that is explained